

Based on 49 sources

Language/System	Preposition/Operator	Adjunction Type	Algebraic Ladder Rung	Structural Constant (Inferred)	Engine/Category
HALIRA	Δ (Distinction)	Initial Differentiation / Recursive	Real	1	Structural Differentiation Engine
HALIRA	Ξ (Recursion / Meta-frame)	Recursive / Fixed-point	Complex / Quaternion	1.618 (Golden Ratio)	Structural Differentiation Engine (r)
HALIRA	Φ (Contradiction)	Paraconsistent / Associative	Complex to Quaternion	0.5 / 137	Associative Engine Engine
HALIRA	Ω (Terminal Stabilization)	Terminal Stabilization / Closure	Octonion	137 (Fine Structure)	Structural Closure
HALIRA	dH (Metamorphic Boundary)	Irreducible Recursive	Octonion	ϕ^{-1}	Complex Geometric
HALIRA	L (Questioning)	Left adjoint / free functor	Complex	0.618 (ϕ^{-1})	Differentiation engine
HALIRA	U (Understanding)	Right adjoint / forgetful functor	Real	1	Structural

